

TASK 01

FUNCTIONAL DEPENDENCIES & ANOMALIES

1. Functional Dependencies (FDs)

From the given data:

a) Patient Details:

- $\text{VisitID} \rightarrow \text{PatientID}, \text{VisitDate}, \text{PatientName},$

b) Doctor Details:

- $\text{VisitID} \rightarrow \text{DoctorID}, \text{DoctorName}, \text{Specialty}, \text{DeptName},$

c) Diagnosis Details:

- $\text{VisitID} \rightarrow \text{Diagnosis}, \text{Fee}$

d) Additional dependencies:

- $\text{PatientID} \rightarrow \text{PatientName}, \text{PatientPhone}$
- $\text{DoctorID} \rightarrow \text{DoctorName}, \text{Specialty}$
- $\text{Specialty} \rightarrow \text{DeptName}, \text{DeptHead}$

2. Anomalies

a) Insertion anomaly:

- Cannot add a new doctor or department unless a visit exists.

b) Update anomaly:

- If doctor's department head changes, multiple rows must be updated.

c) Deletion anomaly:

- Deleting a visit removes patient, doctor, and department information.